

Deep Learning-Based Sewer Pipeline Defect Detection and Severity Analysis Using YOLO

Davala Tejeswara Rao

Department of Electronics and Communication Engineering

Amrita Vishwa Vidyapeetham

Amritapuri, Kollam, Kerala, India

am.en.u4ece24115@am.students.amrita.edu

Abstract—Automated inspection of sewer and drainage pipeline infrastructure is an important requirement for urban asset management and preventive maintenance. This work presents a deep learning-based system for the detection, classification, and severity estimation of sewer pipeline defects observed in closed-circuit television (CCTV) inspection imagery. The proposed system employs YOLOv8m, a single-stage object detection model from the Ultralytics framework, trained on a labelled dataset containing seven defect categories: Buckling, Crack, Debris, Hole, Joint Offset, Obstacle, and Utility Intrusion. Exploratory data analysis reveals pronounced class imbalance and a high prevalence of small bounding-box areas, which represent major detection challenges. The trained model achieves an overall Precision of 0.558, Recall of 0.467, mAP@0.5 of 0.457, and mAP@0.5:0.95 of 0.197 on the validation split. In addition to detection, a rule-based severity analysis module is applied to each predicted defect using normalised bounding-box area, aspect ratio, and defect-specific engineering interpretation. The final system converts raw CCTV inspection frames into structured outputs containing defect class, confidence, and severity level, thereby supporting maintenance prioritisation for pipeline inspection workflows.

Index Terms—sewer pipeline inspection, defect detection, YOLOv8, object detection, severity analysis, CCTV imagery, deep learning, infrastructure maintenance

I. INTRODUCTION

Sewer and drainage pipeline networks are essential components of urban infrastructure, responsible for transporting wastewater and stormwater to treatment and disposal facilities. Over time, these pipelines are exposed to hydraulic stresses, chemical corrosion, soil movement, ageing materials, and external disturbances, which can lead to structural and functional defects. Common sewer pipeline defects include cracks, buckling, holes, joint offsets, debris accumulation, obstacles, and utility intrusions. If such defects are not detected and repaired in time, they may result in pipe collapse, leakage, groundwater contamination, sinkholes, and increased maintenance costs.

Traditional sewer inspection is commonly performed using closed-circuit television (CCTV) cameras, where trained operators manually examine video footage and record observed defects. Although this approach is widely used, it is time-consuming, subjective, and dependent on operator experience. Manual inspection also becomes difficult when large

volumes of inspection footage must be reviewed. Therefore, automated computer vision-based defect detection systems can significantly improve inspection efficiency, consistency, and decision-making.

Recent developments in deep learning have enabled major progress in image classification and object detection. Object detection models are capable of identifying both the category and location of objects within an image. Among these models, the You Only Look Once (YOLO) family has become popular due to its real-time detection capability and favourable balance between accuracy and computational efficiency. This work applies YOLOv8m to the sewer pipeline defect detection problem and further extends the detection output with a defect-specific severity analysis module. Unlike systems that only detect defect classes, the proposed pipeline also estimates the severity of each detected defect, making the output more useful for maintenance prioritisation.

II. RELATED WORK

Early approaches for sewer pipeline defect detection relied on traditional image processing and machine learning techniques. Hand-crafted features such as texture descriptors, edge features, and histograms of oriented gradients were commonly combined with classifiers such as support vector machines. Although these methods provided useful initial results, their performance was limited by sensitivity to illumination changes, camera noise, pipe surface variation, and the difficulty of designing robust features manually.

Deep learning-based object detection methods have improved performance in many visual inspection tasks. Fast R-CNN and Faster R-CNN introduced region-based frameworks for accurate object detection, while the YOLO family reformulated object detection as a single-stage regression problem, allowing faster inference [1]–[3]. Later YOLO versions improved detection accuracy and speed through better feature extraction, multi-scale prediction, and improved training strategies, while residual learning improved deep feature representation in convolutional neural networks [4], [6], [11]. YOLOv8, implemented by Ultralytics, provides an efficient framework for custom object detection tasks and is suitable

for practical inspection applications where both speed and accuracy are important [7].

Recent studies have applied deep learning to drainage and sewer pipeline inspection. Wang *et al.* proposed an enhanced YOLOv8 framework for drainage pipeline defect detection and introduced improvements such as enhanced feature extraction and improved bounding-box regression for complex pipeline defect scenes [9]. Liu *et al.* combined the GLIP pre-trained model with YOLOv5 for water supply and drainage pipeline defect detection, demonstrating the usefulness of pre-trained vision-language models and YOLO-based detectors in urban pipeline inspection [10]. However, many existing studies focus mainly on detection and classification. In contrast, the present work integrates YOLO-based detection with a defect-specific severity analysis module that uses bounding-box geometry and defect-specific engineering reasoning to generate maintenance-priority information.

III. METHODOLOGY

A. Dataset

The dataset used in this work is *Sewer Defects v1i* in YOLO annotation format. It consists of CCTV inspection images of sewer and drainage pipelines annotated with bounding boxes for seven defect classes: Buckling, Crack, Debris, Hole, Joint Offset, Obstacle, and Utility Intrusion. Each annotation file follows the YOLO label format:

$$\text{class_id } x_c \ y_c \ w \ h \quad (1)$$

where (x_c, y_c) represents the normalised bounding-box centre coordinates, and (w, h) represent the normalised width and height of the bounding box. The dataset is divided into training, validation, and test folders, each containing corresponding image and label files.

B. Exploratory Data Analysis

Exploratory data analysis was conducted before training to understand the dataset characteristics. The number of images in the training, validation, and test splits was counted, and bounding-box instances were counted for each class using the YOLO label files. The observed class distribution showed significant imbalance, with Buckling having the highest number of annotated instances and Obstacle having the lowest number of instances. The bounding-box count per class was observed as follows: Buckling: 997, Crack: 201, Debris: 203, Utility Intrusion: 207, Obstacle: 49, Joint Offset: 301, and Hole: 406.

The bounding-box area was calculated as the product of normalised width and height. The area distribution showed that many defects occupy a small region of the image. This is an important challenge because small defects contain fewer visual features and are harder for the model to detect accurately. Random images were also visualised with bounding boxes and class labels to verify annotation correctness. Empty label files were checked and removed to avoid training artefacts.

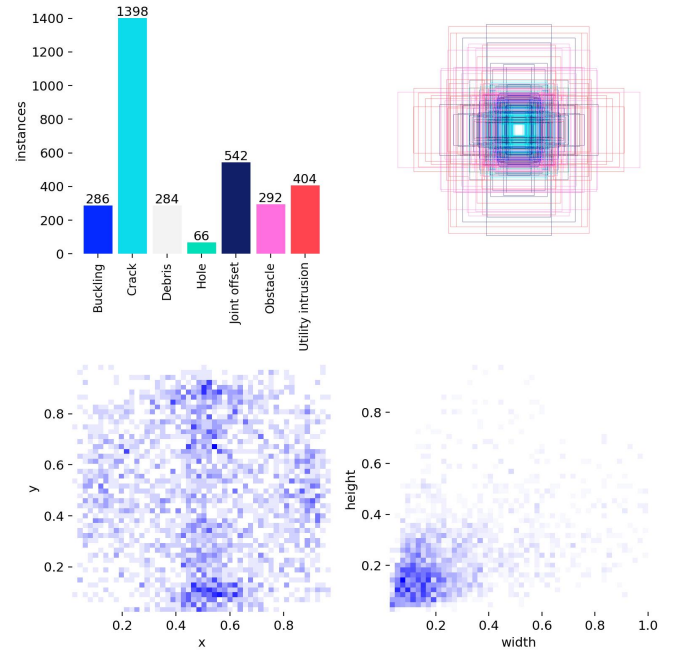


Fig. 1. Dataset annotation overview showing class distribution and bounding-box characteristics.

C. Preprocessing and Data Augmentation

All preprocessing and training were performed in the Google Colab environment. The dataset was extracted from a compressed archive, and the image-label correspondence was verified using matching filenames. Since YOLO internally handles resizing and normalisation during training, no additional manual image normalisation was applied.

Data augmentation was used to improve generalisation and reduce the effect of limited training data. The augmentation methods included horizontal flipping, random brightness and contrast adjustment, rotation, random scaling, mosaic augmentation, and MixUp. Mosaic augmentation combines multiple images into a single training sample and is useful for small-object detection because it increases the diversity of object scales and contexts. MixUp blends image pairs and helps improve model robustness. It is important to note that augmentation does not replace real additional data; rather, it increases the effective diversity of the training samples.

D. Model Architecture and Training Configuration

YOLOv8m, the medium variant of the YOLOv8 model from the Ultralytics framework, was selected for this work. YOLOv8 uses an efficient single-stage detection pipeline with multi-scale feature aggregation, enabling detection of objects at different sizes. The model optimises bounding-box loss, classification loss, and Distribution Focal Loss (DFL) for improved localisation accuracy. Unlike older YOLO versions, YOLOv8 does not rely on a separate explicit objectness loss branch in the same manner.

Training was performed using a Google Colab instance equipped with an NVIDIA Tesla T4 GPU. The model was

trained using pretrained YOLOv8m weights. The input image size was set to 640×640 pixels as a balance between small-object detection capability and GPU memory usage. A batch size of 4 was selected due to memory limitations of the available GPU. The model was trained for approximately 120–150 epochs with early stopping patience to avoid unnecessary training after convergence. Stochastic Gradient Descent (SGD) was used as the optimiser with a learning rate of 0.01 and momentum of 0.937. Training augmentations included mosaic = 1.0, mixup = 0.1, scale = 0.6, rotation of approximately $\pm 8^\circ$, and horizontal flip probability of 0.5.

IV. RESULTS AND DISCUSSION

A. Training Dynamics

The training performance curves show the learning behaviour of the YOLO model over the training epochs. The classification loss decreases over time, indicating that the model progressively learns to assign correct defect categories. The Distribution Focal Loss also decreases, indicating improved bounding-box localisation accuracy. Precision and recall increase during training, showing that the model becomes better at making correct detections and missing fewer actual defects.

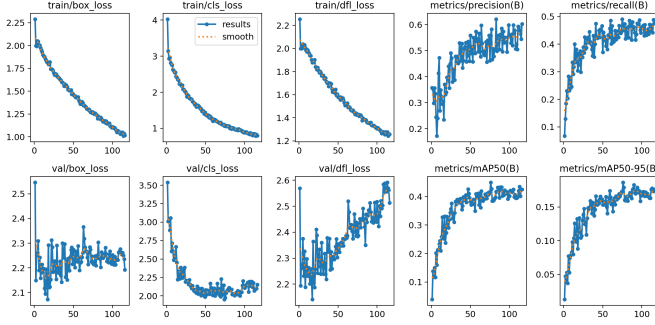


Fig. 2. YOLO training performance curves showing classification loss, Distribution Focal Loss, precision, recall, mAP@0.5, and mAP@0.5:0.95 over training epochs.

The mAP@0.5 curve improves during the early and middle training stages and later begins to stabilise, indicating convergence. The mAP@0.5:0.95 value remains lower than mAP@0.5 because it evaluates detection performance under stricter IoU thresholds. Mild fluctuation in validation metrics suggests slight overfitting, which is expected due to class imbalance, small defect regions, and limited dataset size.

B. Quantitative Performance

Table I presents the per-class and overall validation results. The validation set contains 196 images and 510 annotated instances. The overall model achieves Precision = 0.558, Recall = 0.467, mAP@0.5 = 0.457, and mAP@0.5:0.95 = 0.197.

The results show that Utility Intrusion and Obstacle achieved the best detection performance. Utility Intrusion obtained an mAP@0.5 of 0.740, while Obstacle obtained an mAP@0.5 of 0.622. These classes are visually more distinct

and usually have clearer boundaries, making them easier for the model to detect. Debris also achieved a relatively better mAP@0.5 of 0.525.

In contrast, Joint Offset, Hole, and Crack achieved lower performance. Joint Offset recorded the lowest mAP@0.5 of 0.209, mainly because it is visually subtle and can resemble normal pipe joints. Hole performance was limited by the small number of validation instances. Crack detection is challenging because cracks are often thin, elongated, and low-contrast, making precise localisation difficult. These results indicate that the model performs better on visually distinct defects and struggles with subtle or small defects.

C. Confusion Matrix and Precision–Recall Analysis

The confusion matrix provides insight into correct and incorrect classifications across defect categories. Diagonal values represent correct predictions, while off-diagonal values indicate misclassifications. Confusion is expected between visually similar classes such as Crack and Buckling, as both may appear as elongated surface deformations. Joint Offset may also be confused with background or normal pipe joints due to its subtle visual appearance.

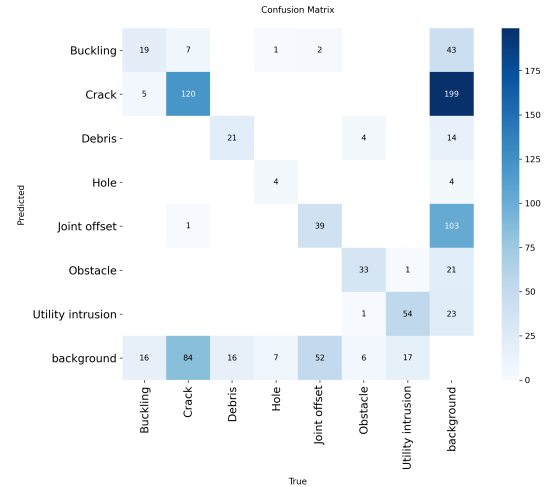


Fig. 3. Confusion matrix evaluated on the validation set.

The normalized confusion matrix provides a class-wise view of prediction behaviour by scaling the values relative to the number of instances in each class. This representation is useful when the dataset is imbalanced, because it allows the performance of minority classes to be interpreted more clearly.

The precision–recall curve evaluates the trade-off between precision and recall at different confidence thresholds. Classes such as Utility Intrusion and Obstacle maintain better precision–recall behaviour, while Joint Offset and Hole show weaker curves due to limited samples and subtle visual features. This analysis confirms that additional data collection, targeted augmentation, and improved small-object detection methods are important directions for future improvement.

The recall–confidence curve shows how recall changes as the confidence threshold is varied. A lower confidence thresh-

TABLE I
PER-CLASS AND OVERALL VALIDATION RESULTS ON THE SEWER DEFECT DATASET

Class	Precision	Recall	mAP@0.5	mAP@0.5:0.95	Images	Instances
Buckling	0.525	0.375	0.366	0.117	34	40
Crack	0.501	0.415	0.381	0.137	66	212
Debris	0.591	0.541	0.525	0.266	32	37
Hole	0.489	0.333	0.356	0.148	10	12
Joint Offset	0.325	0.215	0.209	0.077	40	93
Obstacle	0.719	0.682	0.622	0.263	32	44
Utility Intrusion	0.756	0.708	0.740	0.374	59	72
Overall	0.558	0.467	0.457	0.197	196	510

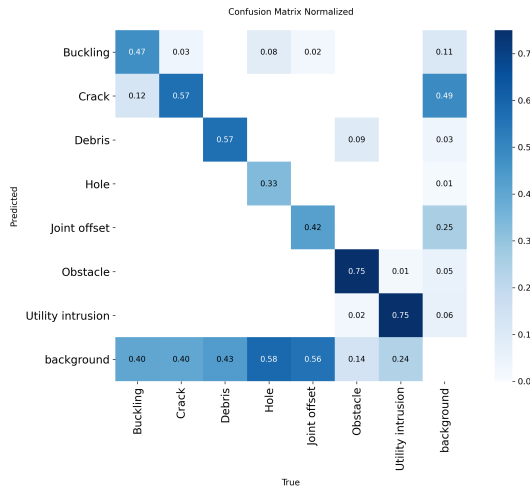


Fig. 4. Normalized confusion matrix evaluated on the validation set.

old generally increases recall but may also introduce more false positives, while a higher threshold improves prediction reliability but may miss subtle defects.

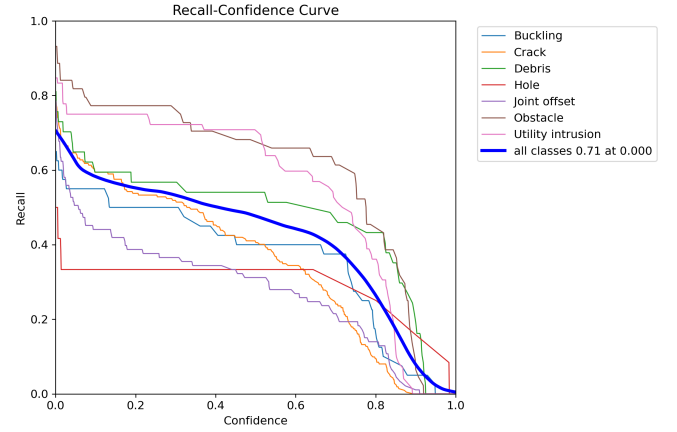


Fig. 6. Recall-confidence curve for bounding-box detection.

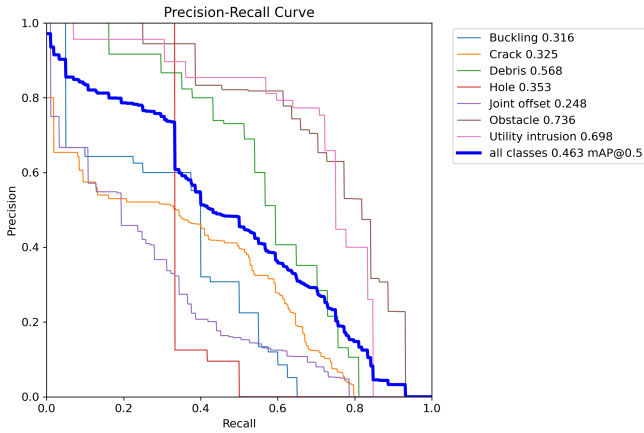


Fig. 5. Precision-recall curve for bounding-box detection across sewer defect categories.

D. Sample Detection Output

A representative YOLO detection output is shown in Fig. 7. The image demonstrates the model's ability to localise sewer pipeline defects using bounding boxes and assign the corresponding defect class labels. This output represents the base object detection stage before applying the severity analysis module.

V. SEVERITY ANALYSIS

A. Severity Module Design

After YOLO-based detection, a rule-based severity analysis module is applied to each detected defect. The module assigns one of three severity labels: *Minor*, *Moderate*, or *Severe*. The severity decision is based on defect class, bounding-box area, and aspect ratio.

The normalised bounding-box area is calculated as:

$$A = w \times h \quad (2)$$



Fig. 7. Representative YOLO detection output showing detected sewer pipeline defects with bounding boxes and class labels.

where w and h are the normalised bounding-box width and height. Area represents the size of the defect. The aspect ratio is calculated as:

$$R = \frac{w}{h} \quad (3)$$

Aspect ratio represents the shape of the defect. This is especially important for cracks because cracks are usually elongated. A long and thin crack may be more severe than a compact defect with similar area.

B. Defect-Specific Severity Rules

Different defects affect pipeline condition in different ways. Therefore, a single generic severity rule is not suitable for all classes. The proposed severity module uses defect-specific rules.

For Crack defects, aspect ratio is used because cracks are elongated. Higher aspect ratio indicates a longer crack and therefore greater structural concern. For Hole defects, area is used because larger holes imply greater leakage risk. For Debris and Obstacle defects, area is used because larger regions indicate greater flow blockage. For Buckling, area represents the extent of deformation. For Utility Intrusion, area indicates the degree of intrusion and obstruction. Joint Offset is treated as *Severe* because it represents structural misalignment between pipe segments.

The thresholds used for area and aspect ratio are heuristic and data-driven. They can be derived from bounding-box area percentiles observed in the dataset and refined using engineering judgement. In future deployment, these thresholds

should be calibrated using real maintenance records and expert inspection outcomes.

C. Practical Importance

The severity analysis module adds practical value to the detection system. Instead of only reporting that a defect is present, the system also provides a severity level that can help inspection engineers prioritise maintenance. For example, a large obstacle or utility intrusion may require urgent attention, while a small isolated defect may be monitored over time. Thus, the proposed system converts raw object detection output into maintenance-priority information.

Fig. 8 shows a representative severity analysis output in which the detected defect is displayed with its predicted class, confidence score, and rule-based severity level. Compared with the base detection output in Fig. 7, the severity-enhanced output provides additional decision-support information for maintenance prioritisation.



Fig. 8. Representative severity analysis output showing detected sewer pipeline defects with class, confidence score, and rule-based severity label.

VI. CONCLUSION

This work presents a YOLO-based sewer pipeline defect detection and severity analysis system for CCTV inspection images. The YOLOv8m model was trained to detect seven defect classes: Buckling, Crack, Debris, Hole, Joint Offset, Obstacle, and Utility Intrusion. Exploratory data analysis showed that the dataset contains class imbalance and many small defect regions, which make the detection task challenging.

The trained model achieved an overall Precision of 0.558, Recall of 0.467, mAP@0.5 of 0.457, and mAP@0.5:0.95 of 0.197. The results indicate moderate detection performance. Utility Intrusion and Obstacle achieved stronger results due

to their visually distinct appearance, while Joint Offset, Hole, and Crack showed lower performance due to subtle features, small size, and limited training samples.

The main contribution of this work is the integration of a defect-specific severity analysis module with YOLO-based detection. The module uses bounding-box area, aspect ratio, and defect-specific engineering reasoning to classify each detected defect as *Minor*, *Moderate*, or *Severe*. This converts detection outputs into maintenance-priority information, making the system more useful for practical sewer inspection workflows.

Future work should focus on increasing the dataset size, collecting more samples for underrepresented classes, applying targeted augmentation, improving small-object localisation, testing higher input resolutions, and validating severity thresholds using real-world maintenance records.

REFERENCES

- [1] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, Las Vegas, NV, USA, 2016, pp. 779–788.
- [2] R. Girshick, "Fast R-CNN," in *Proc. IEEE International Conference on Computer Vision (ICCV)*, Santiago, Chile, 2015, pp. 1440–1448.
- [3] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 39, no. 6, pp. 1137–1149, 2017.
- [4] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, Las Vegas, NV, USA, 2016, pp. 770–778.
- [5] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 42, no. 2, pp. 318–327, 2020.
- [6] C.-Y. Wang, A. Bochkovskiy, and H.-Y. M. Liao, "YOLOv7: Trainable bag-of-freebies sets new state-of-the-art for real-time object detectors," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, Vancouver, Canada, 2023, pp. 7464–7475.
- [7] G. Jocher, A. Chaurasia, and J. Qiu, "YOLO by Ultralytics," Ultralytics Inc., 2023. [Online]. Available: <https://github.com/ultralytics/ultralytics>
- [8] O. Russakovsky *et al.*, "ImageNet large scale visual recognition challenge," *International Journal of Computer Vision*, vol. 115, no. 3, pp. 211–252, 2015.
- [9] J. Wang, A. Sun, C. Wang, X. Zhang, and J. Huang, "Improved deep learning-based detection of drainage pipeline defects using an enhanced YOLOv8 framework," *Journal of Water Process Engineering*, vol. 78, Art. no. 108605, 2025, doi: 10.1016/j.jwpe.2025.108605.
- [10] P. Liu, W. Chen, J. Yu, Q. Zhao, J. Wang, C. Tan, and D. Luo, "Enhancing urban infrastructure: Water supply and drainage pipeline defect detection with the GLIP pre-trained model and the YOLOv5 model," *Journal of Water and Climate Change*, vol. 15, no. 11, pp. 5714–5720, 2024.
- [11] A. Bochkovskiy, C.-Y. Wang, and H.-Y. M. Liao, "YOLOv4: Optimal speed and accuracy of object detection," *arXiv preprint arXiv:2004.10934*, 2020.
- [12] W. Liu, D. Anguelov, D. Erhan, C. Szegedy, S. Reed, C.-Y. Fu, and A. C. Berg, "SSD: Single shot multibox detector," in *Proc. European Conference on Computer Vision (ECCV)*, Amsterdam, Netherlands, 2016, pp. 21–37.